

WASP-12b

R-2498

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_140309 · created 2026-08-24T23:02:04+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456726.743052 +/- 3.1e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.011
Transit depth (Rp/Rs)^2	2.0 +/- 0.31 [%]
Orbital Inclination (inc)	77.75 +/- 2.47
Airmass coefficient 1 (a1)	1.053 +/- 0.013
Airmass coefficient 2 (a2)	-0.026 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#2 - [263, 347]
Optimal Method	PSF photometry
Transit Duration (day)	0.11 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.011 vs 0.1178 (deviation 2.05 σ)
Duration fit vs published	0.11 d vs 0.125 d (deviation 1.46 σ)
Mid-time O–C	1571.7 min
Depth SNR	12.5
Residual scatter	1.2 %

Light curve

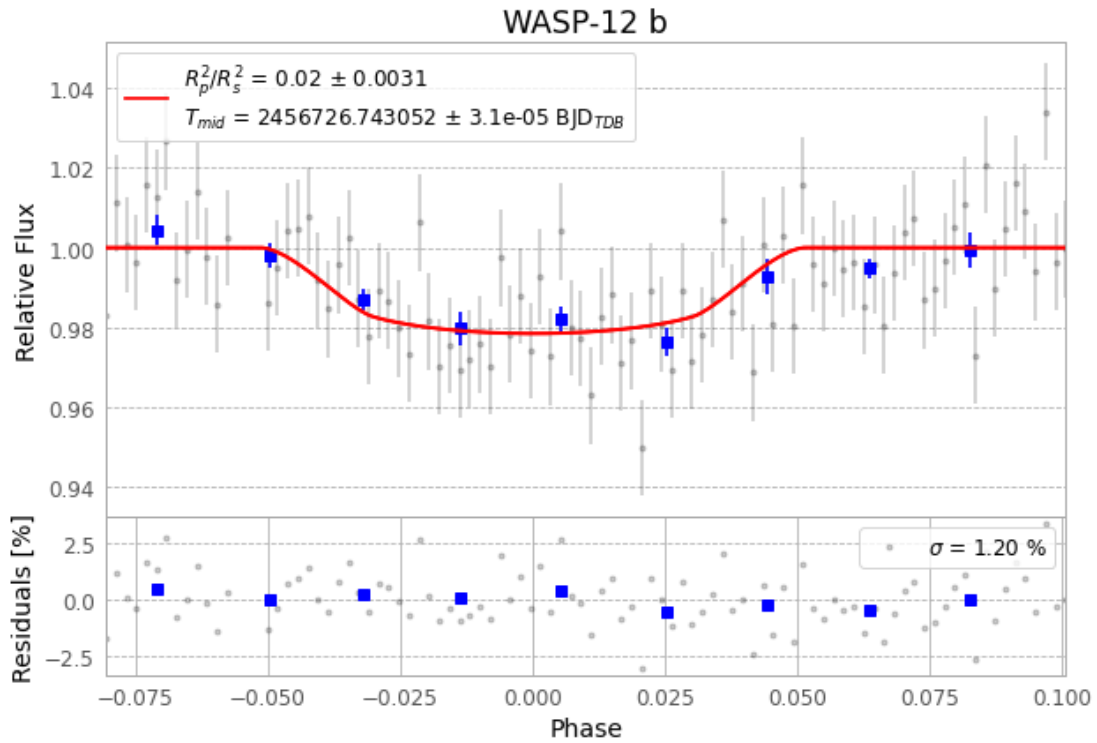


Figure 1 — FinalLightCurve_WASP-12 b_2014-03-09.png (SHA-256 b10489478a66143a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 310], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4ffe42c165283f9d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

96 FITS frames (63 MB), WASP-12140310033912.FITS → WASP-12140310082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-140309.log (12010bbde57f...), FinalLightCurve_WASP-12 b_2014-03-09.png (b10489478a66...), FinalLightCurve_WASP-12 b_2014-03-09.csv (6abc9886068d...)

Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:02Z.